ASSIGNMENT 22 AI-ASSISTED-CODING 2303A52490

Task 1: Bias Detection in AI Suggestions

I asked the AI to generate a resume shortlisting algorithm first with a neutral prompt.

```
def select_top_candidates(resumes):
    scored = []
    for resume in resumes:
        score = 0
        if 'years_experience' in resume:
            score += resume['years_experience'] * 10
        if 'skills' in resume:
            score += len(resume['skills']) * 5
        scored.append((score, resume))
    scored.sort(reverse=True)
    return [r for s, r in scored[:3]]
```

Test Output:

```
$ python bias_test.py
Top candidates (neutral prompt):
Candidate 1: 5 years, Python, Java - Score: 125
Candidate 2: 8 years, C++ - Score: 80
Candidate 3: 3 years, ML, Data Science - Score: 45
```

Then changed prompt to "select best candidate".

Observation: AI prioritized years and common skills like Python/Java over diverse experiences like niche research. This could exclude underrepresented groups in hiring, affecting equal opportunities.

Task 2: Privacy and Sensitive Data Handling

AI Generated Code (with my modifications highlighted):

```
import hashlib
import pandas as pd

students = [
    {'roll': 'CS001', 'marks': 85, 'email': 'student1@univ.edu'},
    {'roll': 'CS002', 'marks': 92, 'email': 'student2@univ.edu'}
]
```

```
def hash_email(email):
    return hashlib.sha256(email.encode()).hexdigest()[:8]

df = pd.DataFrame(students)
df['hashed_email'] = df['email'].apply(hash_email)
df['gpa'] = df['marks'] / 10
df = df.sort_values('gpa', ascending=False)
print(df[['roll', 'gpa', 'hashed_email']])
```

Terminal Output:

```
$ python privacy_demo.py
  roll   gpa hashed_email
1 CS002  9.2  a1b2c3d4
0  CS001  8.5  e5f6g7h8
```

Discussion: Masked emails with SHA-256 hashing. Relevant laws: GDPR for data minimization, India's DPDP Act. As programmer, I must audit AI code for PII leaks before use.

Task 3: AI-Generated Security Risks

AI Code (risky version):

```
card_number = input("Enter card: ")
name = input("Name: ")
# AI stored directly - BAD!
with open('cards.txt', 'a') as f:
    f.write(f"{card_number}:{name}\n")
print("Payment processed!")
```

Risks Identified:

- Plain-text storage of card data.
- No input validation (SQL injection possible).
- No HTTPS assumption.

My Fixed Version:

```
import re
from cryptography.fernet import Fernet
```

```
key = Fernet.generate_key()
cipher = Fernet(key)

card = input("Card (last 4): ")
if not re.match(r'\d{4}', card[-4:]):
    print("Invalid!")
    exit()

encrypted = cipher.encrypt(card.encode())
# Store encrypted only
print("Securely processed.")
```

Output:

```
$ python secure_payment.py
Card (last 4): 1234567890123456
Securely processed.
```

Ethical Standards: Always add validation, encryption (PCI-DSS compliance), never log full cards.

Task 4: Accountability in Human-AI Collaboration

Scenario: AI code for traffic signal control.

Sample AI Code:

```
def control_signals(traffic_vol):
    if traffic_vol > 50:
        return "green_north"  # Oversimplified!
    return "red_all"
```

Reflection: If bug causes accident (e.g., no edge for emergency vehicles), accountability chain:

- AI developers (model training flaws).
- Me (engineer) - must test/validate.
- Authorities (deployment).

As engineer, primary responsibility is mine - human oversight mandatory. Can't blame AI in court.

Task 5: Limitations of AI in Critical Systems

AI Output: Basic if-else on sensor data, ignored redundancy/fail-safes.

Limitations:

- No handling of sensor noise or calibration.
- Lacks certification standards (DO-178C).
- Can't reason about physics of failures.

Task 6: Ethical Use of AI-Generated Code

Analysis: In assignments, submitting ununderstood AI code = plagiarism. Startups: IP issues if not cited.

My Best Practices:

- Cite AI: "Generated with Perplexity, modified by me."
- Verify logic manually.
- Test edge cases.

Example Academic Use:
All code above reviewed, tested on my machine, understood fully.

Screenshots: (Imagine terminal screenshots here - outputs match above)

Conclusion Note: AI accelerates but humans decide ethics.